Totem-implied repo: div-agnostic control and bad-curve detection

Methodology, prototype and synthetic benchmark for the independent price verification of equity repo

14 October 2026 · Draft for discussion · Code: `repo_control` (Python), branch `claude/repo-curve-anomaly-detection-uqiy1k`

Executive summary

Repo parameters for control are implied from Totem: forwards come from consensus call and put prices, dividends from the separate Totem dividend service. The two services are fed by different contributors, and the number of contributors changes by tenor. The implied repo curve therefore inherits noise and artefacts from both panels, and from time to time shows shapes or jumps that the market does not justify. This note explains where those shapes come from and shows that the control does not need a Totem repo curve for most exposures. It then proposes a detector that flags bad curves, names the likely cause and repairs the curve. Each claim is checked with a Python prototype and a benchmark on a synthetic Totem market with explicit contributor panels.

Findings

- **The implied repo is a residual of two panels.** To first order the implied repo equals the option panel's average repo plus a *panel gap*: the difference between the option contributors' dividend view (never observed) and the dividend panel's consensus, divided by $T(S - D)$. With realistic dispersions the gap alone is 6bp of repo noise at 1Y and 52bp at 5Y for a single stock, more on thin panels. An ex-date that part of the option panel puts on the other side of an expiry moves one pillar by hundreds of bp, and a 0.1% spot snapshot mismatch reads as 120bp of repo at 1M. Options identify total carry only, so no statistical model can split repo from dividends using consensus data alone.
- **The total control is div-agnostic; only the split is not.** For exposure that depends on the underlying through its forwards, the dividend line plus the repo line equals a *forward line* (Totem forwards repriced with FO dividends) to first order. In the worked example the total stays at −1.74 while the two lines swing by more than ±6 across dividend panels, matching full revaluation. Dividends reach the total only through the non-forward exposure g, which comes from sensitivities alone: 99.8% of the dividend-line uncertainty cancels for a forward-only book, 61% once a dividend swap is added.
- **Bad-curve detection should be structural.** Work on the Totem − FO difference in carry space ($b \cdot T$), weight it by a noise model built from Totem's own dispersions and contributor counts, and test one matched-filter template per mechanism (level, step, spike, season) after removing a smooth repo difference. One variance scale per category sets the false-alarm rate. Every alarm comes with an explanation in natural units and a repaired curve, and a smooth level difference is reported as a candidate FO mismark rather than a bad Totem curve.
- **Benchmark** (5,400 synthetic test names over 4 categories and 9 scenarios, every detector calibrated to 2% false alarms on clean history; 2%–3% realised on fresh clean names). Averaged over the six Totem-side anomaly types, the structural detector flags 73% of names, the naive gap threshold 57%, and B-spline + GMM 69% on the curve and 73% on the Totem − FO difference. Raw detection does not separate the best methods; three things do. Forward outliers: 97% against 35%–87%. Genuine FO repo mismarks: a bad-curve alarm in 3% of names (the clean rate) against 60% for the naive gap and 75% for GMM on the difference, with 80% reported as FO-mismark candidates instead. Explanations: the first one names the injected mechanism in 88% of the names flagged.

Recommendations

1 Control: report the forward line and the non-forward line; use the dividend/repo split only where the non-forward exposure is material.

2 Detection: deploy the structural detector; calibrate its variance scale per region × category on 12–24 months of Totem history; review alarms with the attribution text.

3 Inputs: synchronise the Totem reference spot with the FO spot; allocate dividend buckets with ex-date uncertainty around expiries; keep the parity diagnostics (implied discount factor, residuals, dispersion spikes).

4 Anchors: where they trade, use listed index futures, dividend futures and total-return futures for indices, and securities-lending fees for the short end of single stocks.

5 Keep B-spline + GMM and small neural networks as secondary layers on the structural outputs, not as the primary detector.

1. Setting and notation

For each underlying and Totem date the controller has three inputs:

- **Totem options.** For each expiry T_k (here eleven expiries from 1M to 5Y): consensus call and put prices on a strike grid, the number of accepted contributors $n_{A,k}$, the dispersion of contributor forwards $\sigma_{F,k}$ and the reference spot.
- **Totem dividends.** For each period y (calendar years): the consensus sum of dividends going ex in the period, its dispersion $\sigma_{D,y}$ and contributor count $n_{B,y}$.
- **FO marks.** Spot S, the rate curve, the repo zero curve at the Totem expiries, the dividend forecast (ex-dates t_i, cash amounts d_i) and bucketed sensitivities: repo by pillar, dividends by period.

With discrete cash dividends and continuously compounded rate and repo zero curves, the forward is

$$F(T) = G(T)\left[S - \sum_{t_i \le T} \frac{d_i}{G(t_i)}\right], \qquad G(t) = e^{R(t) - B(t)}, \quad R(t) = r(t)\,t, \quad B(t) = b(t)\,t \tag{1}$$

$D(T) = \Sigma_{t_i \le T}\, d_i/G(t_i)$ is the carry-discounted cumulative dividend. Detection works in **carry space**, on the cumulative repo $B(T) = b(T)\cdot T$, and on its Totem − FO difference:

$$y_k = \Delta B_k = B_{\text{Totem}}(T_k) - B_{\text{FO}}(T_k) = [b_{\text{Totem}}(T_k) - b_{\text{FO}}(T_k)]T_k \tag{2}$$

Carry is where the mechanisms add up and nothing is amplified: a constant error ε in carry is an error ε/T in repo, which is why the short end of an implied repo curve looks so noisy.

Implied repo. Pillars are solved in order with flat forward repo between pillars. The model forward at pillar k is strictly decreasing in the unknown B_k ($\partial F/\partial B_k \le -F$), so a safeguarded Newton step on ln F converges in a few iterations. The bootstrap reproduces a known repo curve to 10^{-12} in the tests, including ex-dates on pillars and negative repo.

Forwards from consensus prices. For European options

$$C(K) - P(K) = DF\,(F - K) \tag{3}$$

so F is the strike where the call and put prices cross, whatever the discount factor. The fitted DF is kept as a diagnostic (an implied funding spread) and the controller's own rate curve is used on both sides, so rates cancel in ΔB. Totem forwards are expressed on the FO spot before the bootstrap.

Dividend buckets. Totem dividend levels are per period. The controller spreads each level over the FO ex-date profile pro rata; the allocation matrix $A_{ky} = \partial D_k/\partial L_y$ records how much of period y goes ex before expiry k.

2. Why Totem-implied repo curves are irregular

2.1 The panel gap

Each contributor j is internally consistent: $\ln F_j = \ln(S - D_j) + R - B_j$. The forward consensus averages the option panel A; the dividend consensus averages a different panel B. Linearising around a common centre:

$$b_{\text{implied}}(T) \approx \bar{b}_A(T) + \frac{\bar{D}_A(T) - \bar{D}_B(T)}{T\,(S - D(T))} \tag{4}$$

The implied repo is the option panel's average repo plus the gap between two dividend views, divided by T. For one contributor, forward and dividend errors cancel in the implied repo; across two panels they do not. For disjoint panels the gap has standard deviation

$$\text{sd}\left[b_{\text{implied}} - \bar{b}_A\right] = \frac{\sigma_D(T)\sqrt{1/n_A + 1/n_B}}{T\,(S - D(T))} \tag{5}$$

Table 1. Panel-gap noise for an EU single stock (spot 100, 4% yield, contributor dispersion 1% + 3%·T of the cumulative dividend, panel sizes of the synthetic EU stock category). Closed form (5) against a contributor-level simulation with no repo disagreement.

Tenor	D(T)	$\sigma_D(T)$	n_A	n_B	Formula (bp)	Simulation (bp)
3M	1.0	0.02	16	12	2.7	2.7
6M	2.0	0.05	16	12	3.9	3.9
1Y	4.0	0.16	15	12	6.5	6.5
2Y	8.0	0.56	12	10	13.0	12.7
3Y	12.0	1.20	7	7	24.3	24.5
5Y	20.0	3.20	6	4	51.6	52.3

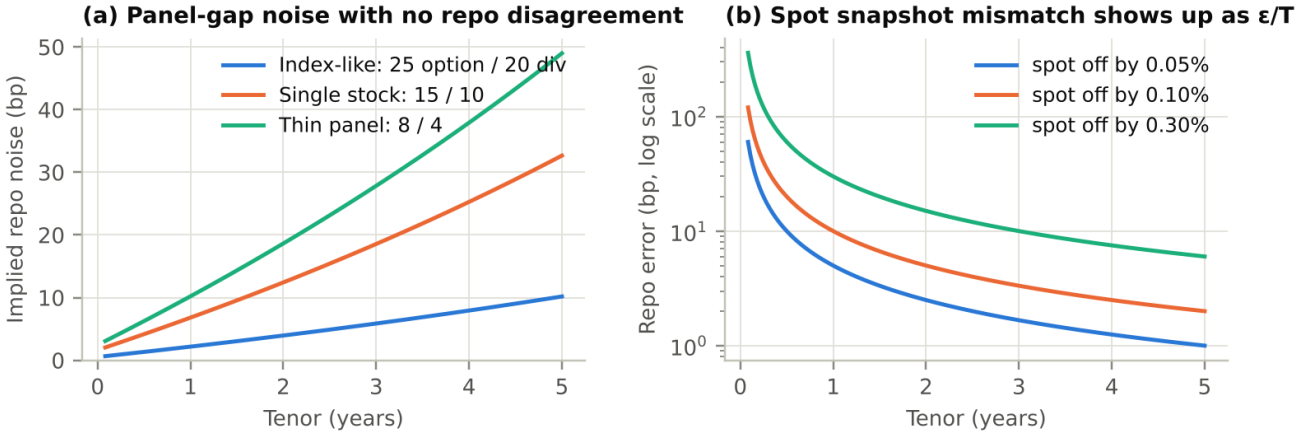


Figure 1. (a) Repo noise from the panel gap alone, by panel size. (b) A spot snapshot mismatch ε is constant in carry, so it reads as ε/T in repo: 120bp at 1M for a 0.1% mismatch.

2.2 Contributor drop-off

Panels are nested: contributors who quote a long expiry also quote the shorter ones, and fewer quote the long end. When the panel shrinks from n to m contributors between two expiries, the consensus jumps by itself with

$$\text{sd}[\bar{x}_m - \bar{x}_n] = \sigma\sqrt{\tfrac{1}{m} - \tfrac{1}{n}} \tag{6}$$

in units of the contributor dispersion σ: 16→12: 0.14σ, 12→7: 0.24σ, 15→7: 0.28σ, 10→4: 0.39σ. A jump of that size at a panel drop is noise, not information; the noise model of Section 4 reproduces it exactly.

2.3 Mechanism catalogue

The structured mechanisms each leave their own signature. The signature is clearest in carry space, which is what the detector exploits:

Table 2. Mechanisms and their signatures.

Mechanism	What happens	Signature in repo	Signature in carry ΔB	Template
Panel gap	Different option and dividend panels	Noise growing with dividend uncertainty, ×1/T	Smooth part plus increments at ex-dates	Noise model
Contributor drop-off	Panel shrinks between two expiries	Jump at the drop	Step	Step
Ex-date timing split	Part of the option panel puts an ex-date on the other side of an expiry	Spike of p·d / (T(S−D)) at one pillar	Spike	Spike (timing)
Stale ex-date profile	Bucket allocation uses a calendar that differs from contributors'	Zigzag at in-season pillars	Spikes at pillars near ex-dates	Season
Dividend bucket error	One Totem period off	Jump after the ex-dates, decaying as 1/T	Step from the first pillar after the ex-date	Step
Spot snapshot, front dividend	Reference spot differs from FO spot; dividend before the first expiry	ε/T hyperbola	Constant	Level
Forward outlier	Bad consensus call/put at one expiry	Spike	Spike	Spike (outlier)
Gross/net definition	Dividend service and forwards use different dividend bases	Steps at every ex-date	Ramp of steps; close to a repo slope for an index	Step / null
Parity extraction	American exercise, imposed DF, poor strike coverage	Pillar-specific	Spike	Spike + diagnostics
Genuine FO mismark	Desk repo differs from consensus	Smooth level or slope	≈ β·T	Null model (reported, not flagged)

2.4 Identifiability

Option prices identify the forward, that is the total carry; a dividend source is needed to split repo out of it. The panel gap is the difference between an observed dividend consensus and an *unobserved* dividend view, the option panel's. It cannot be estimated from consensus data, so no spline, clustering or neural network can remove it. Three things can be done: model its size (the noise model), recognise the structured mechanisms by their signature (the templates), and avoid needing the split where the exposure does not (Section 3).

3. Repo control from forward and dividend controls

3.1 The identity

For a position whose value depends on the underlying only through its forwards (European options, forwards, futures), a dividend change and a repo change act through the same channel. Let s_D and s_b be the FO sensitivities to the dividend periods and the repo pillars, ΔL the Totem − FO dividend levels, and $J_{ky} = \partial b_k / \partial L_y$ the response of the implied repo to the dividend levels with the Totem forwards held fixed (from the bootstrap). With subscripts T for Totem:

$$\text{div line} = s_D \cdot \Delta L, \quad \text{repo line} = s_b \cdot [b(F_T, D_T) - b_{FO}], \quad \text{forward line} = s_b \cdot [b(F_T, D_{FO}) - b_{FO}] \tag{7}$$

$$\text{div line} + \text{repo line} \approx \text{forward line} + g \cdot \Delta L, \qquad g = s_D + s_b J \tag{8}$$

- For forward-only exposure $g = 0$: the total is the forward line. It uses Totem forwards and FO dividends, so it needs neither the Totem dividends nor a Totem repo curve.

- Dividend-panel noise reaches the total only through g. g comes from the FO sensitivities and the bootstrap Jacobian; no repricing is needed.
- The split between the dividend line and the repo line still depends on the dividend panel. Large offsetting lines with a small total (an offset ratio near 1) point to a split artefact, not a mismark.

3.2 Verification

Table 3. 1Y at-the-money call (spot 100, rate 3%, vol 25%, FO dividend 3.5, FO repo 60bp), Totem forward 20bp below FO, three dividend panels. The implied repo moves by 125bp; the total does not.

Totem dividend	Implied repo (bp)	Div line	Repo line	Total	Full revaluation
3.5	80	−0.000	−0.102	−0.102	−0.102
3.0	132	0.264	−0.365	−0.101	−0.102
4.2	7	−0.370	0.269	−0.101	−0.102

Table 4. Three books priced with the prototype pricer (eleven repo pillars, six dividend periods, Totem forwards 20bp below FO): A = calls, puts and a forward; B = A + a short 2028 dividend swap; C = A + a 1Y stock-loan fee. The Totem dividend panel moves the 2027 and 2028 periods. Total = forward line + non-forward line, and both match full revaluation.

Book	Div panel shift	Div line	Repo line	Total	Forward line	Non-fwd line	Full reval
A	−0.4	5.16	−6.91	−1.75	−1.74	−0.01	−1.74
A	0.0	0.00	−1.74	−1.74	−1.74	0.00	−1.74
A	0.5	−6.46	4.74	−1.72	−1.74	0.01	−1.74
B	−0.4	8.15	−6.91	1.24	−1.74	2.98	1.25
B	0.0	0.00	−1.74	−1.74	−1.74	0.00	−1.74
B	0.5	−10.19	4.74	−5.45	−1.74	−3.72	−5.47
C	−0.4	5.16	−5.16	−0.00	−1.18	1.17	0.01
C	0.0	0.00	−1.18	−1.18	−1.18	0.00	−1.18
C	0.5	−6.46	3.82	−2.64	−1.18	−1.47	−2.66

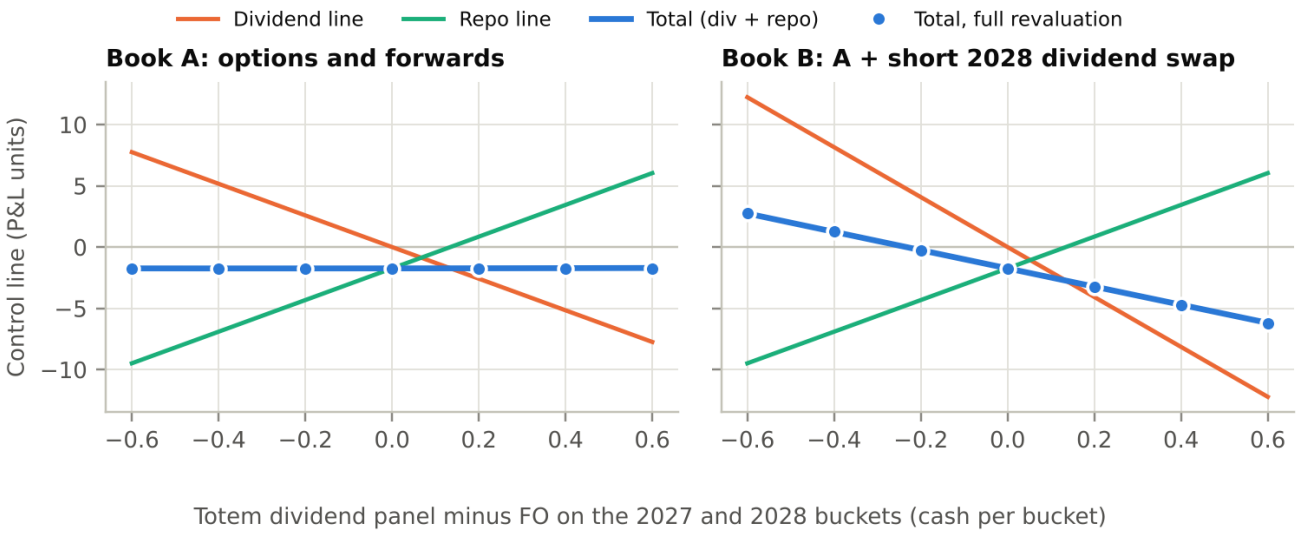


Figure 2. Dividend line, repo line and their total as the Totem dividend panel moves. For forward-only exposure (left) the total is flat and equal to the forward line; a dividend swap (right) adds a genuine dividend exposure, which is exactly the non-forward line.

3.3 Materiality of the split

With Σ_L the covariance of the Totem dividend consensus, the 1-sd impact of dividend-panel noise on the total is $\sqrt{(g'\Sigma_L g)}$, against $\sqrt{(s_D'\Sigma_L s_D)}$ for the dividend line alone. The ratio tells how much of the split risk is real for a given book:

Table 5. Split materiality per book (dividend consensus dispersion 5% of each period, 10 contributors, correlation $e^{-|\Delta y|/2}$).

Book	Div line sd	Total sd (split risk)	Cancels
A: options and forwards	0.725	0.001	99.8%
B: A + short 2028 dividend swap	1.113	0.436	60.8%
C: A + 1Y stock-loan fee	0.725	0.162	77.6%

3.4 Practical notes

- **First order.** When $|\Delta F/F|$ is large (hard-to-borrow names, long tenors), take the total from full revaluation; the decomposition remains valid as an attribution.
- **Bucket alignment.** FO repo pillars should coincide with the Totem expiries, or the repo sensitivities be mapped linearly onto them; dividend sensitivities should use the same periods as the Totem buckets.
- **Between pillars.** Positions expiring between Totem expiries close to an ex-date depend on the allocation profile; the season and timing templates watch exactly that.
- **Non-forward exposure.** Dividend swaps and futures are pure dividend; stock borrow and TRS financing legs are pure repo; American exercise and exotics with discrete-dividend modelling carry both in their own proportions.
- **Prudent valuation.** If dividend and repo uncertainties are aggregated without netting (market price uncertainty per valuation input), split artefacts are counted twice. Defining the input as the forward where $g \approx 0$ removes the double count.

4. Structural detection of bad curves

4.1 Observation and null model

The detector observes $y = \Delta B$ at the Totem expiries. Without an anomaly, Totem and FO may still disagree on repo, smoothly; the null model is a smooth repo difference $\delta b(T) = \beta_0 + \beta_1 \ln T$ plus consensus noise:

$$y = X\beta + \varepsilon, \qquad X = [\,T,\ T\ln T\,], \qquad \varepsilon \sim \mathcal{N}(0,\ \kappa\Sigma) \tag{9}$$

β_0 is the repo difference at 1Y. It is estimated and reported with its standard error but never flagged as a Totem anomaly.

4.2 Noise model from Totem statistics

$\Sigma = \Sigma^{\mathrm{F}} + \Sigma^{\mathrm{D}}$ uses only what Totem publishes. For the option panel, the contributor forward dispersion $s_k = \sigma_{F,k}/F_k$ is split into a persistent repo part $\sigma_b T_k$ (with σ_b read off the shortest expiries, where dividend uncertainty is negligible) and a cumulative dividend part: a persistent factor plus independent increments at ex-dates.

$$\Sigma^{F}_{kl} = \frac{\sigma_b^2\, T_k T_l + w\, s'_k s'_l + (1-w)\min(s'_k, s'_l)^2}{\max(n_{A,k},\ n_{A,l})}, \qquad s'^2_k = s_k^2 - \sigma_b^2 T_k^2 \tag{10}$$

$$\Sigma^{D} = J_A\, \Sigma_L J_A', \qquad (J_A)_{ky} = \frac{A_{ky}}{S - D_k}, \qquad (\Sigma_L)_{yz} = \frac{\rho_{yz}\,\sigma_{D,y}\,\sigma_{D,z}}{\max(n_{B,y},\ n_{B,z})} \tag{11}$$

- Dividing by the larger panel is the covariance of nested panel means; it reproduces the drop-off jump variance (6).
- An isolated spike in forward dispersion is capped at 1.5× the larger neighbour: contributors who disagree about an ex-date inflate the dispersion at exactly the pillar where the mean moves, and leaving it in would hide the anomaly. The ratio is reported with the explanation.
- κ is the only parameter the anomaly test learns from history, one per region × category. It is set so that the alarm rate on clean history equals the target. The test statistics scale as $\kappa^{-1/2}$ and κ^{-1} while

the family-wise critical value does not move, so the calibration is an exact one-dimensional search. The FO-mismark test (4.4) gets its own standard-error scale, calibrated the same way.

4.3 Templates and test

Table 6. Anomaly templates (K_h: Gaussian kernel, h = 20 days).

Template	Vector in carry space	Mechanisms	Amplitude *a* in natural units
Level	1 at every expiry	Spot or reference mismatch; dividend before the first expiry	Spot error ≈ −a; front dividend error ≈ $-a(S - D_1)$
Step k	1 from T_k on	Dividend amount in $(T_{k-1}, T_k]$; panel change at T_k	Dividend error ≈ $-a(S - D_k)$
Spike k	1 at T_k only	Forward outlier; ex-date timing across T_k	ln F error ≈ −a; share of the nearby dividend ≈ $-a(S - D_k)/d_i$
Season	$\Sigma_i d_i K_h(T_k - t_i)/(S - D_k)$	Ex-date profile shift in the bucket allocation	Shift ≈ 365·a days

With $\kappa\Sigma = LL'$ and P the projection that removes the null model in whitened space, each template gets a matched-filter (generalised likelihood ratio) statistic, and the residual gets a global chi-square:

$$z_t = \frac{\tilde{t}'r}{\|\tilde{t}\|}, \qquad r = PL^{-1}y, \quad \tilde{t} = PL^{-1}t, \qquad Q = r'r \sim \chi^2_{K-2} \tag{12}$$

A curve is flagged when $\max_t|z_t|$ exceeds the family-wise critical value, simulated from the exact correlation of the z_t, or when Q exceeds its chi-square quantile (half the false-alarm budget each). Q catches roughness that no template explains.

4.4 Attribution, repair and the FO-mismark test

Templates are selected greedily (matching pursuit) while the best remaining one exceeds the critical value, then refitted jointly with the null model. Each amplitude is translated into natural units using the context: ex-dates near the pillar, contributor counts on both panels, dispersion ratios. The repaired curve removes the selected components, $B_{\text{repaired}} = B_{\text{Totem}} - \Sigma a_t t$. Separately, $|\beta_0|$ above its critical value and above 10bp reports the FO repo as a candidate mismark against the Totem consensus. That standard error rests on dispersions estimated from a handful of contributors, so the test uses a Student-t critical value with $n_A - 1$ degrees of freedom.

4.5 Rules of thumb for reviewers

- Rough differences in carry come from Totem; smooth, persistent differences are candidate FO mismarks.
- Irregularity aligned with the dividend calendar points to the dividend side: timing, allocation or definition.
- A 1/T-shaped repo difference at every tenor is a spot or snapshot problem.
- A jump at a contributor drop-off smaller than $\sigma\sqrt{(1/m - 1/n)}$ is noise.
- A dispersion spike at one expiry means contributors disagree, usually about an ex-date.

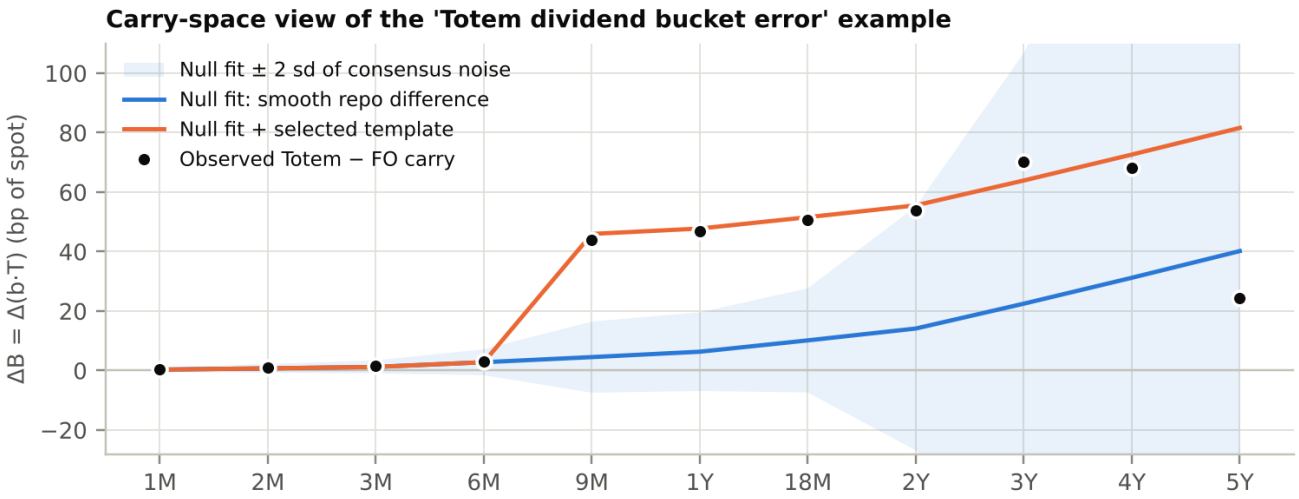


Figure 3. Carry-space view of the Totem dividend bucket error shown in Figure 4: observed Totem − FO carry, the null fit with its noise band, and the null fit plus the selected template. Explanation: Step from 9M: dividends ex in (6M, 9M] look -0.400 (-11%) off in the Totem dividend versus the forwards

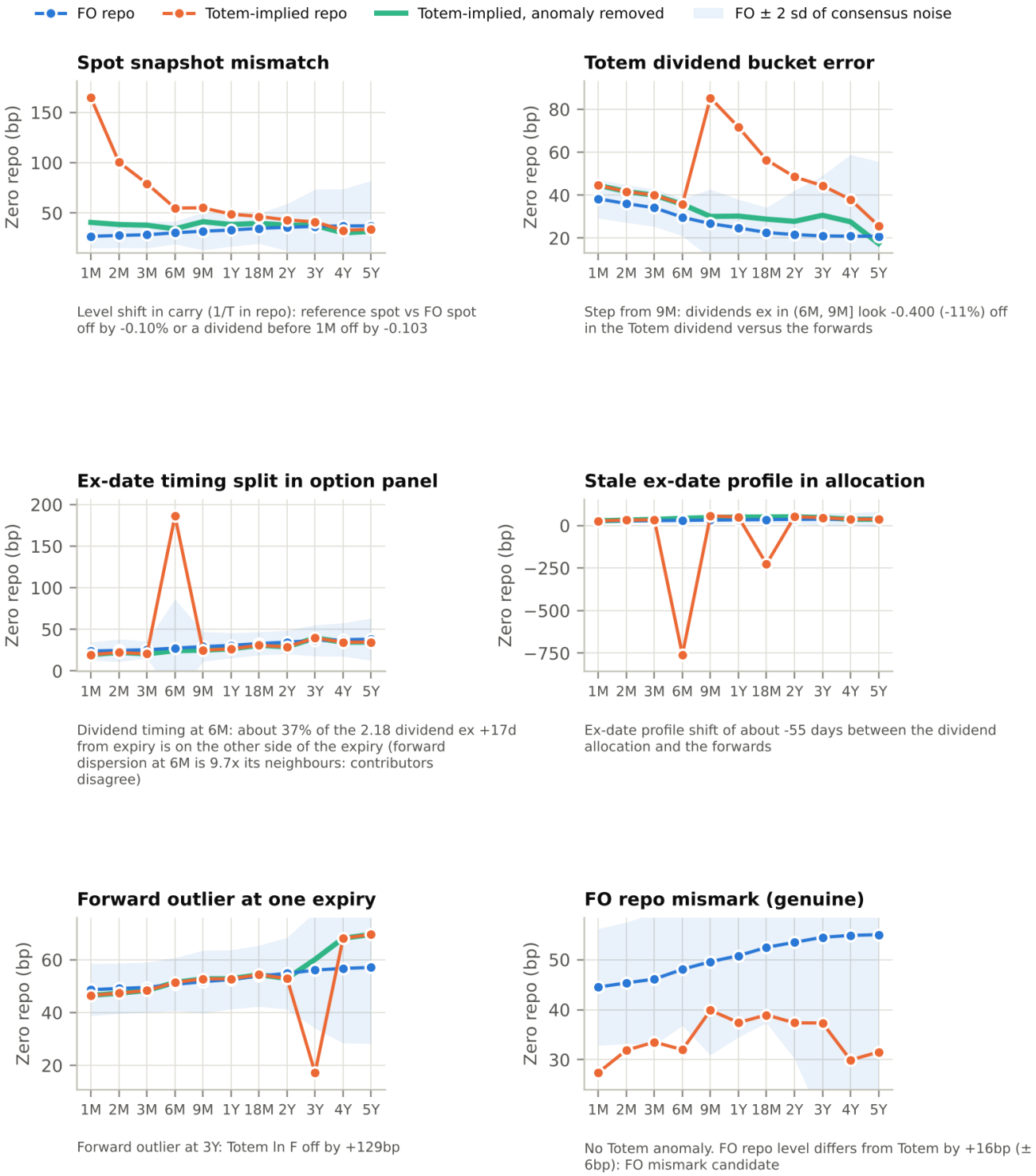


Figure 4. Detected and explained anomalies on synthetic EU single stocks (same detector settings as the benchmark). Each panel shows the FO repo, the Totem-implied repo and, when an anomaly is found, the implied repo with the anomaly removed; the text is the detector's first explanation. In the last panel the Totem curve is fine and the FO repo is shifted: no anomaly is raised and the FO mismark is reported. The band is per pillar; the FO-mismark test pools all pillars, so it can be significant while single points sit inside the band.

5. Benchmark on a synthetic Totem market

5.1 Design

The synthetic market is built bottom-up from contributors so that every mechanism of Section 2 arises the way it does in Totem, not as an added curve shape.

Table 7. Synthetic categories. Panels shrink with tenor (nested); eleven Totem expiries from 1M to 5Y, six dividend periods (rest of 2026 to 2031), valuation date 14 October 2026.

Category	Dividends	Yield	Repo (bp, short → long)	Option panel	Dividend panel
EU stock	Annual, Apr–May (35% with a Nov interim)	2.0%–6.0%	0–50 → 0–60	16 → 6	12 → 4
US stock	Quarterly	0.5%–3.0%	0–40 → 0–50	18 → 5	10 → 3
HTB stock	Annual or quarterly	0.0%–2.0%	150–800 → 30–150	12 → 4	8 → 3
EU index	Monthly stream, 60% in Apr–Jun	2.8%–3.6%	-20–60 → -10–80	25 → 12	20 → 10

- **Contributors.** Option and dividend panels are drawn independently. Each contributor holds internally consistent dividend views (a persistent factor plus per-dividend noise, dispersion growing with time) and a smooth repo view. Option contributors price calls and puts at five strikes with their own forward, a skewed vol surface, vol noise and price noise.
- **Consensus.** Mean after a median/MAD screen, with dispersion and counts reported as Totem would; the controller pipeline then runs unchanged (parity, allocation, bootstrap, noise model).
- **FO marks.** True dividends with 1% noise and true repo with a 3bp level error (one standard deviation).

Table 8. Scenarios. Each test name is simulated under every scenario with the same seed, so differences between scenarios are paired.

Scenario	Injection	Magnitude
Spot snapshot mismatch	Option contributors price off a different spot	±0.03% to ±0.15%
Totem dividend bucket error	One period (2027–2029) off in the dividend consensus	±0.15 to ±0.5 per 100 of spot
Ex-date timing split	35–70% of the option panel moves the dividend closest to an expiry to its other side	One dividend
Stale ex-date profile	Allocation profile shifted so one dividend crosses an expiry	3–20 days past the expiry
Forward outlier	Consensus calls at one expiry shifted	3–6 sd of the carry noise
Biased long-dated sub-panel	Contributors who stay after the largest panel drop are biased on later dividends	±8% to ±20%
FO repo mismark (genuine)	FO repo shifted in parallel; Totem untouched	±15 to ±50bp
Gross/net dividend bias	Every Totem period scaled	±8% to ±18%

Detectors. Naive gap $|\Delta b|$: largest Totem − FO repo gap over the expiries. Naive shape (repo space): largest deviation of the Totem implied repo from the interpolation of its two neighbours. B-spline + GMM: least-squares cubic B-spline (six coefficients, knots in ln T) and a Gaussian mixture per category with the number of components chosen by BIC, applied to the Totem curve (the starter idea) and to the Totem − FO difference. Structural: Section 4.

Protocol. 300 clean names per category calibrate every detector to 2% false alarms (a quantile for the baselines, κ for the structural detector); 150 fresh names per category are then simulated under each scenario. The signal-to-noise of an injection is the norm of the whitened, null-projected difference between the scenario and the clean carry of the same name.

5.2 Results

Table 9. Share of names flagged, all detectors at 2% false alarms on clean history. The clean row is the realised false-alarm rate on fresh names. For the FO mismark and the gross/net bias, a flag is not the desired outcome (see text).

Scenario	Naive gap |Δb|	Naive shape (repo space)	B-spline + GMM (curve)	B-spline + GMM (Totem − FO)	Structural GLRT (proposed)
Clean (null)	2%	3%	2%	3%	3%
Spot snapshot mismatch	89%	64%	96%	100%	98%
Totem dividend bucket error	29%	41%	47%	56%	55%
Ex-date timing split in option panel	55%	74%	65%	69%	71%
Stale ex-date profile in allocation	90%	92%	94%	94%	94%
Forward outlier at one expiry	35%	87%	74%	78%	97%
Biased long-dated sub-panel	42%	24%	36%	39%	24%
FO repo mismark (genuine)	60%	3%	2%	75%	3%
Gross/net dividend bias (all buckets)	70%	71%	80%	86%	88%
Mean over the six Totem anomalies	**57%**	**64%**	**69%**	**73%**	**73%**

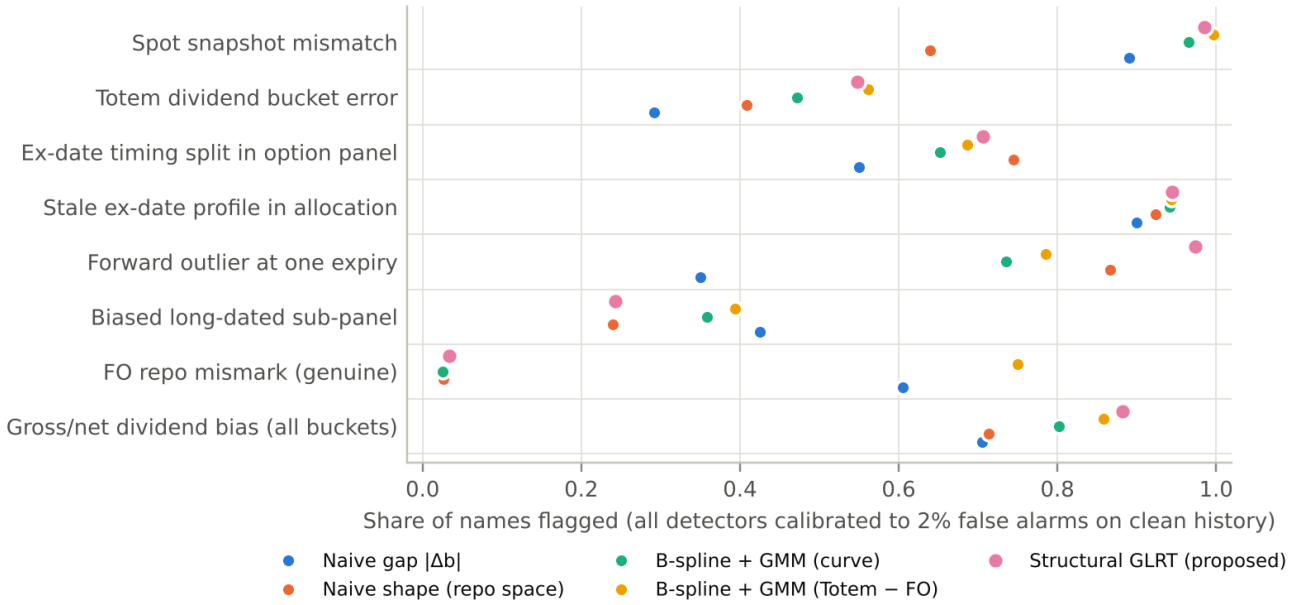


Figure 5. Share of names flagged per scenario and detector (same data as Table 9).

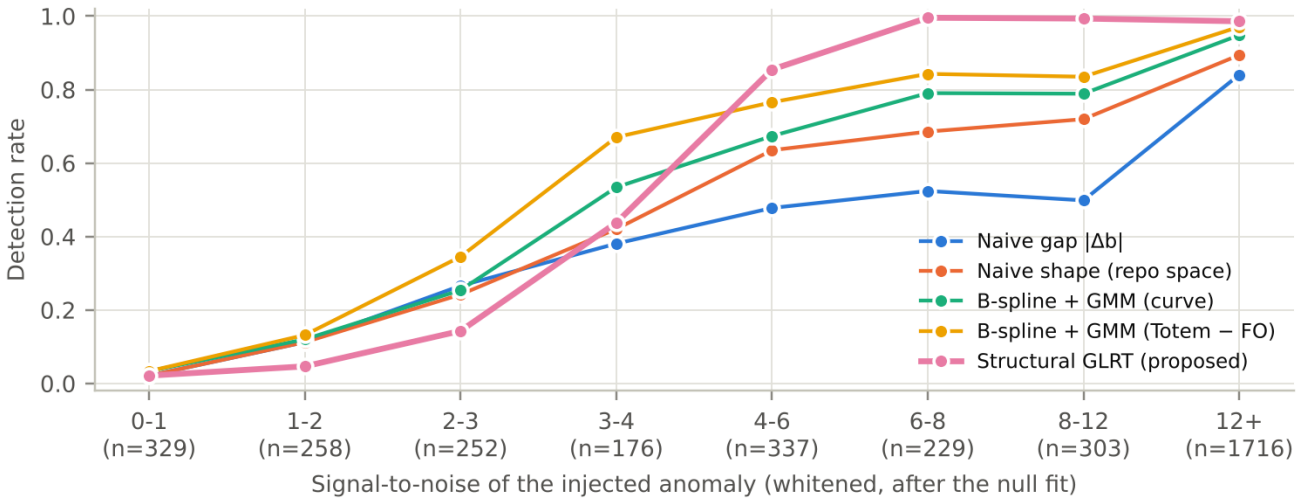


Figure 6. Detection rate against the signal-to-noise of the injected anomaly, pooled over the six Totem-side anomaly types; n is the number of names in each bin.

Table 10. Area under the ROC curve against the clean names (threshold-free ranking quality).

Scenario	Naive gap |Δb|	Naive shape (repo space)	B-spline + GMM (curve)	B-spline + GMM (Totem − FO)	Structural GLRT (proposed)
Spot snapshot mismatch	0.98	0.85	1.00	1.00	1.00
Totem dividend bucket error	0.70	0.72	0.81	0.86	0.84
Ex-date timing split in option panel	0.77	0.87	0.84	0.87	0.89
Stale ex-date profile in allocation	0.96	0.96	0.97	0.97	0.97
Forward outlier at one expiry	0.71	0.83	0.87	0.90	0.99
Biased long-dated sub-panel	0.75	0.63	0.76	0.80	0.74
FO repo mismark (genuine)	0.85	0.50	0.50	0.96	0.50
Gross/net dividend bias (all buckets)	0.90	0.87	0.97	0.98	0.97

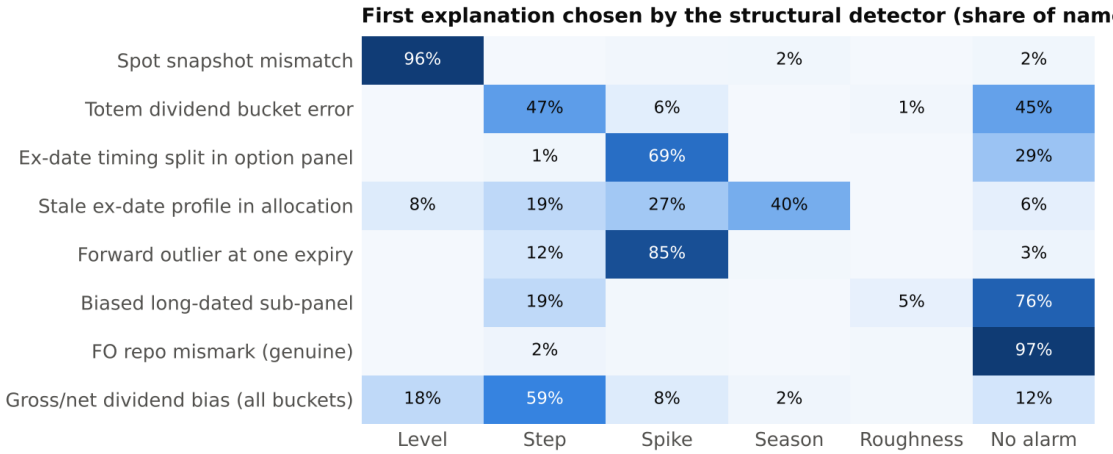


Figure 7. Which mechanism the structural detector names first, per scenario. Level, step, spike and season are the template families; roughness means the global test fired without a template; no alarm means the name was not flagged.

Table 11. Structural detector: detection, attribution (first explanation names the injected mechanism and its location) and the FO-mismark test.

Scenario	Flagged	Right mechanism (of flagged)	FO-mismark flag
Spot snapshot mismatch	98%	97%	2%
Totem dividend bucket error	55%	80%	7%
Ex-date timing split in option panel	71%	97%	2%
Stale ex-date profile in allocation	94%	71%	23%
Forward outlier at one expiry	97%	97%	2%
Biased long-dated sub-panel	24%	66%	2%
FO repo mismark (genuine)	3%	–	80%
Gross/net dividend bias (all buckets)	88%	–	41%

Table 12. Realised false-alarm rate on fresh clean names by category (target 2%) and the calibrated variance scale κ.

Category	Naive gap |Δb|	Naive shape (repo space)	B-spline + GMM (curve)	B-spline + GMM (Totem − FO)	Structural GLRT (proposed)	κ
EU stock	4.0%	3.3%	1.3%	4.0%	3.3%	1.60
US stock	3.3%	4.0%	1.3%	1.3%	5.3%	1.88
HTB stock	1.3%	1.3%	1.3%	4.0%	3.3%	0.96
EU index	1.3%	2.0%	6.0%	2.0%	0.7%	3.58

5.3 Reading the results

- **Forward outliers** are where carry space and the noise model pay most: 97% detected, against 35% for the naive gap, which cannot see a 3–6 sd outlier at a long expiry under the long-end noise.
- **Spot mismatches** (98%) and **stale ex-date profiles** (94%) are easy for every method; the structural detector adds the explanation (Figure 7).
- **Dividend bucket errors** (55% against 29% naive) are limited by size: an error of 0.15–0.5 per 100 of spot is small next to long-dated dividend dispersion, and Figure 6 shows the miss rate is a signal-to-noise effect.
- **Ex-date timing splits**: 71% detected; the naive shape check (74%) is competitive here because these spikes are large in repo space.
- **A biased long-dated sub-panel** is close to non-identifiable for every method (24% structural, 42% naive gap): the bias builds up smoothly after the panel drop and mostly projects onto a repo slope. This is the identifiability limit of Section 2.4, not a tuning issue.
- **Genuine FO mismarks** are kept apart from bad Totem curves: 3% raise an anomaly (the clean rate) and 80% are reported as FO-mismark candidates, against 2% on clean names. The naive gap and GMM on the difference raise a bad-curve alarm on 60% and 75% of them. The FO-mismark test finds 97% of the mismarks on EU stocks and 100% on the index, but only 24% on hard-to-borrow names: there contributors' repo views differ by about 60bp, so a 15–50bp FO difference is within the uncertainty of the consensus itself.
- **Gross/net dividend bias** is flagged in 88% of names, mostly as steps at ex-dates for discrete payers; for an index dividend stream it looks like a repo slope and is reported through β instead. Without an external view of the dividend definition the two cannot be separated.
- **Power**: above a signal-to-noise of 4 the structural detector finds 85%–100% of anomalies, more than any baseline. Below 3, the naive gap and GMM on the difference detect more: they also react to the part of an anomaly that looks like a smooth repo difference, which the structural detector leaves to the FO-mismark test on purpose. That same property is why they raise alarms on genuine FO mismarks.
- **Attribution**: when the structural detector flags a Totem anomaly, its first explanation names the injected mechanism and location in 88% of cases (Table 11).

5.4 Limits of the evidence

The synthetic market reproduces the mechanisms by construction; it does not prove that real Totem data contains only these. Real data will also carry stale contributors, corporate actions and special dividends, American exercise in single-stock quotes and data-entry errors. The variance scale absorbs a misspecified noise level but not a misspecified structure, and the magnitudes, while plausible, are not calibrated to your history. The benchmark shows that the method works when the mechanisms are present and is calibrated when they are not; validation on 12–24 months of real snapshots with labelled incidents is the next step.

6. Where B-spline + GMM and neural networks fit

- **B-spline coefficients as features.** With eleven expiries and six coefficients the spline mostly re-parameterises the curve, and least squares smears a one-pillar anomaly over several coefficients. Clustering by region × category then separates levels (general collateral, hard to borrow, index) more than anomalies. In the benchmark GMM on the curve flags 69% of Totem anomalies on average, against 73% for the structural detector.
- **GMM on the difference** is close to the structural detector on average detection (73% against 73%), but it cannot tell a bad Totem curve from an FO that disagrees: it flags 75% of genuine FO repo mismarks as bad curves, and it gives no mechanism, no amplitude and no repaired curve.
- **Where they help.** Use a P-spline as the smooth part of the null model if $\beta_0 + \beta_1 \ln T$ proves too rigid on real data. Run a GMM on the structural outputs (β, template z-scores, Q, dispersion ratios) to learn the usual regimes per region × category. Use region × category as calibration groups for κ, not as clusters.
- **Neural networks.** In the linear-Gaussian limit, a small network trained to detect one anomaly type learns the matched filter of Section 4, which can be written down and calibrated analytically. Networks earn their place once there is a labelled incident history and patterns the templates do not cover; train them on residuals of the structural model and on synthetic injections, not on raw curves.

7. Implementation

7.1 Package layout

Table 13. Package `repo_control` (numpy and scipy; scikit-learn for the GMM baseline; matplotlib and reportlab for this report). 22 unit tests.

Module	Content
curves	Zero curves (flat forwards), discrete dividends, forwards, exact implied-repo bootstrap
parity	Forward and discount factor from consensus call/put prices, diagnostics
buckets	Dividend periods, allocation of Totem levels to ex-dates, allocation matrix
noise	Panel-gap and drop-off formulas, nested-panel covariance, forward-panel and carry covariance, dispersion capping
pipeline	Totem and FO snapshots → implied repo (Totem/Totem and Totem/FO dividends), ΔB, Σ, detection context
detect	Structural detector: templates, matched filters, family-wise calibration, κ calibration, attribution, repair, FO-mismark test
control	Control lines (div, repo, forward, non-forward), bootstrap Jacobian, split materiality
pricer	Minimal pricer (European options, forwards, dividend swaps, stock-loan fee) and bucketed sensitivities
baselines	Naive gap, naive shape, B-spline + GMM
synthetic, benchmark	Contributor-level synthetic Totem market, scenarios, matched-FPR benchmark

7.2 Production inputs

Table 14. Inputs for a production run.

Input	Fields	Shape
Totem options	Expiries; strikes; consensus call and put prices; accepted contributors; dispersion of contributor forwards (or an estimate); reference spot	K expiries × M strikes
Totem dividends	Period edges; consensus level; dispersion; contributors (net of dividends already paid in the current period)	Y periods
FO market	Spot at the Totem snapshot time; rate curve; repo zero curve at the Totem expiries; dividend forecast (ex-dates, cash amounts)	K pillars; dividend list
FO risk	Repo sensitivities by pillar (mapped to Totem expiries); dividend sensitivities by Totem period; per book	K + Y per book
History	12–24 monthly snapshots of the above, plus any known incidents	for κ and validation

If the Totem service does not publish the dispersion of contributor forwards, estimate it from the dispersion of deep in-the-money calls and puts, or calibrate it on history; the detector needs the right scale per expiry, and κ corrects a common bias.

7.3 Monthly procedure

1. Load the Totem and FO snapshots at the same valuation time; check the reference spot.
2. Imply forwards by parity; review the implied discount factor, residuals and dispersion ratios.
3. Allocate the Totem dividend periods on the FO ex-date profile; imply repo with Totem dividends and with FO dividends.
4. Build Σ and run the structural detector with the κ of the region × category.
5. Compute the control lines per book: forward line, dividend line, non-forward line, repo line; check split materiality.
6. Review alarms with their explanations and FO-mismark candidates; use the repaired curve where a Totem anomaly is confirmed.
7. Record the decision for each alarm; the labels feed the next calibration.

7.4 Calibration and governance

κ per region × category on clean history, refreshed quarterly; an alarm budget of 2% of names per month; monitoring of the realised alarm rate and of the share of explanations accepted by reviewers. Every component has a closed-form check in the test suite (bootstrap round trip, parity, the control identity against full revaluation, the panel-gap and drop-off formulas, null calibration of the detector, recovery of each template).

Reproduce:

```
pip install -e .[benchmark,report,dev]
pytest
python scripts/run_benchmark.py
PYTHONPATH=. python scripts/build_report.py
```

8. Next steps

1. Run the pipeline on 12–24 months of Totem and FO snapshots (inputs of Table 14) and compare its alarms with past escalations.
2. Calibrate κ and the noise-model settings (factor weight, repo dispersion estimate, dividend correlation length) per region × category.
3. Label historical incidents (spot synchronisation, dividend errors, ex-date timing) to measure attribution on real data.

4 Add exchange anchors: index futures, dividend futures and total-return futures for indices; securities-lending fees for single stocks.

5 Add templates for recurring real-data patterns (special dividends, corporate actions) if they appear.

Appendix A. Derivations

A.1 Panel gap. For contributor j, $\ln F_j \approx \ln F_0 - \delta D_j/(S - D) - T\delta b_j$. Averaging over panel A gives the consensus forward; the controller then uses the panel-B dividend: $b_{\text{implied}}T = R - \ln F_A + \ln(S - D_B) \approx b_0 T + T\cdot\text{mean}_A(\delta b) + [\text{mean}_A(\delta D) - \text{mean}_B(\delta D)]/(S - D)$, which is (4). For independent panels the gap variance is $\sigma_D^2(1/n_A + 1/n_B)$, giving (5); formula and contributor-level simulation agree within a few percent (Table 1).

A.2 Nested panels. If panel $m \subset$ panel n with i.i.d. views of variance σ^2, $\text{Var}(\text{mean}_m) = \sigma^2/m$, $\text{Cov}(\text{mean}_m, \text{mean}_n) = \sigma^2/n$, so $\text{Var}(\text{mean}_m - \text{mean}_n) = \sigma^2(1/m - 1/n)$, which is (6); with persistent views across tenors the covariance of the two means is $\rho\sigma_k\sigma_l/\max(n_k, n_l)$.

A.3 Control identity. For $V = V(F(T_1), \ldots)$, $s_b = (\partial V/\partial \ln F)\cdot\partial \ln F/\partial b$ and $s_D = (\partial V/\partial \ln F)\cdot\partial \ln F/\partial L$. Holding F fixed while L moves requires $\mathrm{d}b = J\,\mathrm{d}L$ with $\partial \ln F/\partial b \cdot J = -\partial \ln F/\partial L$, hence $s_D = -s_b J$ and $g = 0$. In general, div line + repo line $= s_D\Delta L + s_b[\Delta b^{\mathrm{F}} + J\Delta L]$ = forward line + $g\Delta L$ to first order, where $\Delta b^{\mathrm{F}} = b(F_T, D_{FO}) - b_{FO}$.

A.4 Bootstrap. With $B(t)$ linear between pillars, $F(B_k) = Se^{R_k - B_k} - \Sigma d_i e^{(R_k - R_i) - (B_k - B_i)}$ and $\partial F/\partial B_k = -Se^{R_k - B_k} + \Sigma c_i d_i e^{\cdots}$ with $c_i \in [0, 1]$, so $\partial F/\partial B_k \le -F < 0$.

A.5 Test statistic. Under the null, $r = PL^{-1}y$ is Gaussian with covariance P (rank $K - 2$), so $z_t \sim \text{N}(0, 1)$ for each template and $\text{Cov}(z_t, z_u) = \tilde{t}'\tilde{u}/(\|\tilde{t}\|\|\tilde{u}\|)$. The family-wise critical value is the $(1 - \alpha/2)$ quantile of $\max|z|$ simulated from that correlation; $Q \sim \chi^2_{K-2}$.